

Describing Meaning Handout

Week 9: Thematic Roles, Event Structure, and Experiencer Predicates

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Useful reminders

1. Some common role labels

Role	Rough description
AGENT	volitional, controlling participant
CAUSER	participant that brings about an event or change, without necessarily being volitional
THEME	participant that moves, changes, is located, or is emitted
PATIENT	participant affected by an event
GOAL	endpoint of motion or change
SOURCE	starting point of motion or change; with some emission verbs, the emitting entity
PATH	route of motion, without necessarily specifying source or goal
RECIPIENT	possessor reached in transfer
BENEFACTIVE	intended possessor / beneficiary
EXPERIENCER	participant that feels, perceives, or mentally represents something
STIMULUS	participant that is perceived or that provokes an emotion

2. LCS-style reminders

Some useful LCS patterns:

agent = x in $DO(x, [\dots])$

causer = x in x CAUSE $[\dots]$

theme = y in $BECOME[predicate(y)]$

goal = z in $BECOME[be-at(y, z)]$

object-experiencer pattern = $x \text{ CAUSE } [\text{BECOME}[feel(y, state)]]$

3. Neo-Davidsonian Semantics

In Neo-Davidsonian notation, the verb introduces an event variable, and thematic roles are predicates of that event.

John broke the window.

$$\exists e(\text{break}(e) \wedge \text{Agent}(e, j) \wedge \text{Theme}(e, w))$$

The storm broke the window.

$$\exists e(\text{break}(e) \wedge \text{Causer}(e, s) \wedge \text{Theme}(e, w))$$

Sarah threw the branch at the window.

$$\exists e(\text{throw}(e) \wedge \text{Agent}(e, s) \wedge \text{Theme}(e, b) \wedge \text{Goal}(e, w))$$

Some event predicates:

$\text{Agent}(e, x), \text{Causer}(e, x), \text{Theme}(e, x), \text{Goal}(e, x), \text{Source}(e, x), \text{Path}(e, x),$

$\text{Experiencer}(e, x), \text{Stimulus}(e, x), \text{Recipient}(e, x), \text{Benefactive}(e, x)$

Exercise: Neo-Davidsonian notation

For each sentence, identify the thematic roles of the DPs then translate it into Neo-Davidsonian notation.

- a. The vase broke.
- b. The faucet dripped water onto the floor.
- c. Selena sent Luke a postcard.
- d. Maya baked Leo a cake.
- e. The article enraged Thomas.
- f. Someone was enraged by the article.
- g. All of the hikers trudged along the ridge to the lodge from their camp.

Verb classes and alternations

- what **class of verb** are we dealing with?
- what **alternations** does this verb participate in?
- which semantically similar verbs **fail** to alternate in the same way?

Compare the following:

- a. *Jones gave a bone to the puppy.*
Jones gave the puppy a bone.
Jones donated a bone to the shelter.
**Jones donated the shelter a bone.*
- b. *Selena sent the report to her manager.*
Selena sent her manager the report.
Selena described the report to her manager.
**Selena described her manager the report.*
- c. *Selena sprayed paint on the wall.*
Selena sprayed the wall with paint.
Selena poured paint on the wall.
**Selena poured the wall with paint.*

Exercise: LCS and alternations

- a. Lucinda wove the twigs into a wreath.
Lucinda wove a wreath from the twigs.
 - b. Cheryl cleared the dishes from the table.
Cheryl cleared the table of dishes.
 - c. Selena sprayed paint on the wall.
Selena sprayed the wall with paint.
 - d. John broke the vase.
The vase broke.
 - e. Jones gave a bone to the puppy.
Jones gave the puppy a bone.
- Which participant undergoes the most salient change in each sentence?
 - Does one variant entail the other?
 - Does the alternation preserve meaning completely?

Exercise: From LCS to Neo-Davidsonian notation

For each LCS below:

- (i) give an English verb that could lexicalise it;
- (ii) identify the main thematic roles;
- (iii) give a Neo-Davidsonian representation using that verb.

- a. x CAUSE [BECOME[*broken*(y)]]
- b. $DO(x, [walk(x)])$
- c. x CAUSE [BECOME[*be-at*(y, z)]]
- d. x CAUSE [BECOME[*not-be-at*(y, z)]]
- e. x CAUSE [BECOME[*feel*($y, angry$)]]

Psych-verb puzzles

A major challenge for simple role inventories comes from experiencer predicates.

Compare:

- a. Midge loves ice cream.
- b. Jones feared the poltergeist.
- c. Ruth enjoyed the cable car ride.
- d. The movie bored Simon.
- e. The article enraged Thomas.
- f. The television set worried John.

At first glance, all of these seem to involve an experiencer and something like a stimulus. But the syntactic patterns differ:

- in some cases the experiencer is the **subject**;
- in others the experiencer is the **object**;
- the non-experiencer participant is not always doing the same semantic job.

A Reinhart-style way of thinking about psych predicates

One useful way of thinking about these contrasts due to Tanya Reinhart is in terms of two features:

- [+c] = causes / brings about change
- [+m] = mental state / sentience

Role	Cause	Mental state
Agent	+	+
Causer / Instrument	+	–
Experiencer	–	+
Theme / Patient	–	–

The idea is that these form a hierarchy where when two positions are available the higher one will be occupied by the higher role in the table and vice versa.

Exercise:

For the following experiencer sentences.

- (i) Is the non-experiencer DP more like a cause or more like the target of the experience?
- (ii) Do the patterns of realisation track Reinhart's categorisation?

- a. The article enraged Thomas.
- b. The television set worried John.
- c. Midge loves ice cream.
- d. Jones feared the poltergeist.
- e. The poltergeist frightened John.
- f. The movie bored Simon.
- g. Jones believed that oysters were poisonous in May.

Now consider the sentences below? Do they fit Reinhart's pattern?

- a. The test worries John.
- b. It worries John that the test is tomorrow.
- c. John worries that the test is tomorrow.
- d. That the test is tomorrow worries John.